

## Paper 2 - Set 11 - Solutions

Generated practice paper: answer key and worked solutions.

### Answer key

| Question | Answer | Topic                                   |
|----------|--------|-----------------------------------------|
| 1        | B      | Quadratic functions                     |
| 2        | F      | Simultaneous equations and inequalities |
| 3        | C      | Indices and surds                       |
| 4        | B      | The logic of arguments                  |
| 5        | A      | Mathematical proof                      |
| 6        | A      | Identifying errors in proofs            |
| 7        | D      | Mathematical proof                      |
| 8        | A      | Identifying errors in proofs            |
| 9        | H      | Graphs of functions                     |
| 10       | E      | Trigonometric functions and equations   |
| 11       | E      | Exponentials and logarithms             |
| 12       | F      | Graphs of functions                     |
| 13       | E      | Chains of reasoning                     |
| 14       | E      | Trigonometric functions and equations   |
| 15       | H      | Chains of reasoning                     |
| 16       | G      | The logic of arguments                  |
| 17       | F      | Quadratic functions                     |
| 18       | A      | Mathematical proof                      |
| 19       | E      | The logic of arguments                  |
| 20       | C      | Differentiation                         |

## Worked solutions

### Question 1 QUA-07 · Quadratic functions · spec MM1.3

How many real solutions does the equation  $x^4 - 5x^2 + 4 = 0$  have?

**Answer: B**

Let  $u = x^2$ :  $u^2 - 5u + 4 = (u - 1)(u - 4) = 0$ , so  $x^2 = 1$  or  $x^2 = 4$ , giving  $x = \pm 1, \pm 2$ : four solutions.

#### Full method - how and why

A quartic that is quadratic in  $x^2$ . The trap is counting the solutions of the substituted equation rather than of the original.

1 Substitute  $u = x^2$ :  $u^2 - 5u + 4 = 0$ .

why: Only even powers of  $x$  appear, which is what makes the substitution work.

2 Factorise:  $(u - 1)(u - 4) = 0$ , so  $u = 1$  or  $u = 4$ .

why: Two values of  $u$  - but  $u$  is not the answer.

3 Undo the substitution:  $x^2 = 1$  gives  $x = \pm 1$ ;  $x^2 = 4$  gives  $x = \pm 2$ .

why: Each **positive** value of  $u$  produces **two** values of  $x$ ; a negative value would produce none.

4 Count: four real solutions.

why: The question asks how many, not what they are.

#### Where it goes wrong

- Answering 2 because the quadratic in  $u$  has two roots.
- Forgetting the negative square roots and answering 2 again.

**Takeaway** After a substitution, always translate back: each positive value of  $x^2$  gives two values of  $x$ , zero gives one, negative gives none.

### Question 2 SIM-06 · Simultaneous equations and inequalities · spec MM1.4

Solve  $xy = 6$  and  $x - y = 1$ . What is the sum of all possible values of  $x + y$ ?

**Answer: F**

$x = y + 1$ :  $y^2 + y - 6 = 0 \Rightarrow (y + 3)(y - 2) = 0$ . Solutions (3, 2) with  $x + y = 5$  and  $(-2, -3)$  with  $x + y = -5$ . The sum is 0.

#### Full method - how and why

A product and a difference. Substituting the linear relation gives a quadratic with two solution pairs, and the question wants a total over both.

1 From  $x - y = 1$ , write  $x = y + 1$ .

why: Substituting the linear equation keeps things to one variable.

2 Substitute into  $xy = 6$ :  $(y + 1)y = 6 \Rightarrow y^2 + y - 6 = 0$ .

why: Expand before rearranging into standard form.

3 Factorise:  $(y + 3)(y - 2) = 0$ , so  $y = -3$  or  $y = 2$ .

why: Both are valid - nothing in the question restricts the sign.

4 Find the pairs:  $y = 2 \Rightarrow x = 3$ , giving  $x + y = 5$ ;  $y = -3 \Rightarrow x = -2$ , giving  $x + y = -5$ . Sum of these: 0.

why: The question asks for the sum of **all possible values of  $x + y$** , so compute  $x + y$  for each pair and add.

#### Where it goes wrong

- Finding only the positive pair and answering 5.
- Adding all four numbers  $3 + 2 + (-2) + (-3)$  - which happens to be 0 too, but for the wrong reason; on another question that luck runs out.

**Takeaway** When a system has several solution pairs, compute the requested quantity for each pair and then combine.

### Question 3 IND-15 · Indices and surds · spec MM1.1

Given that  $2^x = 5$  and  $2^y = 20$ , find the value of  $2^{2x-y}$ .

**Answer: C**

$$2^{2x-y} = \frac{(2^x)^2}{2^y} = \frac{25}{20} = \frac{5}{4}.$$

#### Full method - how and why

You are given two exponentials and asked for a third. Express the target in terms of the given ones using index laws - you never need to find  $x$  or  $y$ .

- 1 Split the index:  $2^{2x-y} = \frac{2^{2x}}{2^y}$ .

why: Subtraction in the index is division - this is what lets you use the two given facts separately.

- 2 Write  $2^{2x} = (2^x)^2 = 5^2 = 25$ .

why: A coefficient in the index is a power of the whole exponential:  $2^{2x} = (2^x)^2$ , not  $2 \times 2^x$ .

- 3 Substitute:  $\frac{25}{20} = \frac{5}{4}$ .

why:  $2^y = 20$  is given directly.

#### Where it goes wrong

- Using  $2^{2x} = 2 \times 2^x = 10$ , giving  $\frac{1}{2}$ .
- Trying to solve for  $x = \log_2 5$  and  $y = \log_2 20$  first - correct but slow, and it invites arithmetic slips.

**Takeaway** Rewrite the target index as sums and multiples of the given ones; substitution then beats solving for the unknowns.

### Question 4 LGC-17 · The logic of arguments · spec Arg1

Let  $x$  be a real number. Which of the following statements is equivalent to ' $x \geq 2$  if and only if  $x^3 \geq 8$ '?

**Answer: B**

'If and only if' is the conjunction of the implication and its converse.

#### Full method - how and why

Unpacking 'if and only if' into its two implications.

- 1 'A if and only if B' means both 'A if B' ( $B \Rightarrow A$ ) and 'A only if B' ( $A \Rightarrow B$ ).

why: The phrase is literally built from 'if' and 'only if' - each supplies one direction.

- 2 With  $A: x \geq 2$  and  $B: x^3 \geq 8$ , that is  $(x \geq 2 \Rightarrow x^3 \geq 8)$  **and**  $(x^3 \geq 8 \Rightarrow x \geq 2)$ .

why: The conjunction of the implication and its converse.

- 3 (Both happen to be true here, since cubing is increasing - but the question only asks for the equivalent form.)

why: Read what is asked: the structure, not the truth value.

#### Where it goes wrong

- Giving only one implication - that is half the statement.
- Joining the two with 'or' instead of 'and'.

**Takeaway**  $A \Leftrightarrow B$  is exactly  $(A \Rightarrow B)$  **and**  $(B \Rightarrow A)$ .

### Question 5 X2-27 · Mathematical proof · spec Prf1

Consider the claim: **if  $n$  is a positive integer then  $n^2 + n + 1$  is odd**. Which of the following is a valid proof?

**Answer: A**

The argument from  $n^2 + n = n(n + 1)$  being even is direct and general. Checking  $n = 1, 2, 3$  is only cases; 'cannot be factorised' is irrelevant to parity; 'if  $n$  is odd ...' says nothing about even  $n$ ; and the contradiction attempt never reaches a contradiction, since  $n$  may well be even.

#### Full method - how and why

Same 'which is a valid proof' pattern as PRF-01: general argument versus cases, irrelevance and incompleteness.

- 1 The valid proof:  $n^2 + n = n(n + 1)$  is a product of consecutive integers, hence even; adding 1 gives an odd number.

why: General, covering every positive integer with no case split.

- 2 Checking  $n = 1, 2, 3$  gives 3, 7, 13 - evidence, not proof.

why: Three cases out of infinitely many.

- 3 'It cannot be factorised, so it is odd' - factorisability has nothing to do with parity ( $6 = 2 \times 3$  and  $9 = 3 \times 3$  are respectively even and odd).

why: An irrelevant premise cannot support the conclusion.

- 4 'If  $n$  is odd then ... so  $n^2 + n + 1$  is odd' - correct but only for odd  $n$ ; even  $n$  is untreated. Incomplete.

why: A case-based proof must cover every case.

- 5 'Suppose  $n^2 + n + 1$  is even. Then  $n$  is even, contradiction' -  $n$  being even is not a contradiction;  $n$  can be even.

why: A contradiction proof must actually reach an impossibility.

#### Where it goes wrong

- Accepting the odd- $n$  argument because every line in it is true - truth is not completeness.
- Accepting the contradiction attempt because it uses the right vocabulary.

**Takeaway**  $n(n + 1)$  is always even, so  $n^2 + n + 1$  is always odd. A proof must be general, relevant and complete.

**Question 6** ERR-10 · Identifying errors in proofs · spec Err2

Claim: if  $a > b$  then  $\frac{1}{a} < \frac{1}{b}$ .

I Suppose  $a > b$ .

II Dividing both sides by  $ab$ :  $\frac{1}{b} > \frac{1}{a}$ .

III So  $\frac{1}{a} < \frac{1}{b}$ .

Which of the following is correct?

**Answer: A**

$a = 1, b = -1$ :  $a > b$  but  $\frac{1}{a} = 1 > -1 = \frac{1}{b}$ . Dividing by  $ab$  without knowing its sign is invalid (and  $b = 0$  would be division by zero).

**Full method - how and why**

The same unknown-sign trap as ERR-05, now with division, plus a false claim.

1 Test the claim:  $a = 1, b = -1$ . Then  $a > b$  but  $\frac{1}{a} = 1$  and  $\frac{1}{b} = -1$ , so  $\frac{1}{a} > \frac{1}{b}$ . **Claim false.**

why: Opposite signs are the natural test for a reciprocal claim.

2 Line I is just the supposition - fine.

why: No deduction yet.

3 Line II divides by  $ab$  and keeps the direction. That needs  $ab > 0$ , which is not given;  $ab$  could be negative (as in the counterexample) or zero. **First error.**

why: Dividing by a negative reverses the inequality, and dividing by zero is undefined.

4 The claim is true if  $a$  and  $b$  have the same sign - which is the condition that was silently assumed.

why: Naming the missing hypothesis is the cleanest way to describe this kind of error.

**Where it goes wrong**

- Accepting the argument because it 'works for positive numbers' - the claim is stated for all reals.
- Blaming line III, which is a trivial restatement of line II.

**Takeaway** Reciprocals reverse order only within a fixed sign: for  $a, b > 0$  (or both negative),  $a > b \Rightarrow \frac{1}{a} < \frac{1}{b}$ .

### Question 7 PRF-07 · Mathematical proof · spec Prf1

Prove by cases that  $n^3 - n$  is divisible by 3 for every integer  $n$ . Which set of cases is both sufficient and necessary to consider?

**Answer: D**

$n^3 - n = (n - 1)n(n + 1)$  is a product of three consecutive integers; splitting by remainder on division by 3 shows one factor is a multiple of 3 in every case.

#### Full method - how and why

Proof by exhaustive cases: the cases must cover every integer (sufficient) with none redundant (necessary), and they must be chosen to match the divisor in question.

**1** The claim is about divisibility by 3, so split by remainder on division by 3:  $n = 3k, 3k + 1, 3k + 2$ .

*why:* Every integer has exactly one of these three forms, so the cases are exhaustive and non-overlapping.

**2** Check each:  $n^3 - n = (n - 1)n(n + 1)$ . If  $n = 3k$  the middle factor is a multiple of 3; if  $n = 3k + 1$  then  $n - 1 = 3k$ ; if  $n = 3k + 2$  then  $n + 1 = 3(k + 1)$ .

*why:* Three consecutive integers always include a multiple of 3 - the case split makes that explicit.

**3** So the product is divisible by 3 in every case.

*why:* The proof is complete because the cases cover all integers.

#### Where it goes wrong

- Splitting into even/odd - that is the wrong modulus and tells you nothing about divisibility by 3.
- Using only  $3k + 1$  and  $3k + 2$ , which omits the multiples of 3 themselves - not exhaustive.
- Splitting by sign or by primality: exhaustive, but useless for this claim.

**Takeaway** For divisibility by  $m$ , split into the  $m$  residue classes  $mk, mk + 1, \dots, mk + (m - 1)$ .

### Question 8 ERR-18 · Identifying errors in proofs · spec Err2

A student solves  $2\cos^2 x = \cos x$  for  $0 \leq x < 2\pi$ .

I Divide both sides by  $\cos x$ :  $2\cos x = 1$ .

II So  $\cos x = \frac{1}{2}$ .

III So  $x = \frac{\pi}{3}$  or  $x = \frac{5\pi}{3}$ .

How many solutions has the student missed, and where is the first error?

**Answer: A**

Dividing by  $\cos x$  loses  $\cos x = 0$ , i.e.  $x = \frac{\pi}{2}$  and  $x = \frac{3\pi}{2}$ : two solutions.

#### Full method - how and why

Dividing a trigonometric equation by  $\cos x$  loses the solutions where  $\cos x = 0$ .

1 Line I divides by  $\cos x$ , valid only if  $\cos x \neq 0$ . **First error.**

*why:* The equation  $2\cos^2 x = \cos x$  is satisfied whenever  $\cos x = 0$ , so those solutions are discarded.

2 The correct method:  $2\cos^2 x - \cos x = 0$ , so  $\cos x(2\cos x - 1) = 0$ .

*why:* Factorise rather than divide - it keeps both branches.

3 Branch  $\cos x = 0$ : in  $0 \leq x < 2\pi$  that is  $x = \frac{\pi}{2}$  and  $x = \frac{3\pi}{2}$  - **two** missed solutions.

*why:* Cosine vanishes at the top and bottom of the unit circle.

4 Branch  $\cos x = \frac{1}{2}$ :  $x = \frac{\pi}{3}$  and  $x = \frac{5\pi}{3}$  - the two the student found.

*why:* Cosine is positive in the first and fourth quadrants, so  $x = \frac{\pi}{3}$  and  $2\pi - \frac{\pi}{3}$ . Four solutions in total.

#### Where it goes wrong

- Saying one solution was missed;  $\cos x = 0$  has two solutions in a full revolution.
- Blaming line III - the student's own two values are correct for their (incomplete) equation.
- Declaring the working correct because the two values found do satisfy the original equation.

**Takeaway** Never divide a trigonometric equation by a trig function - factorise, and count the solutions of **each** branch across the whole interval.

**Question 9** GRA-05 · Graphs of functions · spec MM8.1

Which of the following statements about the graph of  $y = |2x - 4|$  is/are true?

- I** It meets the  $x$ -axis at exactly one point.
- II** It is symmetric about the line  $x = 2$ .
- III** It lies entirely on or above the line  $y = 0$ .

**Answer: H**

$|2x - 4| = 0$  only at  $x = 2$  (I).  $|2x - 4| = 2|x - 2|$  is symmetric about  $x = 2$  (II). A modulus is never negative (III).

**Full method - how and why**

Three properties of a modulus graph. Rewriting  $|2x - 4|$  as  $2|x - 2|$  makes all three obvious.

- 1** Rewrite:  $|2x - 4| = 2|x - 2|$ .  
*why:* Factoring the 2 out of the modulus is legitimate because  $|2u| = 2|u|$ .
- 2** I: the graph meets  $y = 0$  when  $|x - 2| = 0$ , i.e. only at  $x = 2$ . **True.**  
*why:* A V-shape touches the axis at its single vertex.
- 3** II:  $|x - 2|$  depends only on the distance from 2, so the graph is symmetric about  $x = 2$ . **True.**  
*why:* Replacing  $x$  by  $4 - x$  leaves the value unchanged.
- 4** III: a modulus is never negative, so the graph lies on or above the  $x$ -axis. **True.**  
*why:*  $|u| \geq 0$  by definition.
- 5** All three.  
*why:* Match to the option list.

**Where it goes wrong**

- Thinking the graph crosses the axis twice because  $|x^2 - 4|$ -type graphs do - this one is linear inside the modulus.
- Reading the line of symmetry as  $x = 4$ .

**Takeaway**  $|ax + b| = |a||x + \frac{b}{a}|$ : a V-shape with its vertex at the root, never below the axis.

**Question 10** X2-07 · Trigonometric functions and equations · spec MM4.6

Consider the equation  $\sin x = k$  where  $k$  is a real constant. Which of the following statements is/are true?

- I For every  $k$  with  $-1 < k < 1$  the equation has exactly two solutions in  $0 \leq x < 2\pi$ .
- II For  $k = 1$  the equation has exactly one solution in  $0 \leq x < 2\pi$ .
- III For every real  $k$  the equation has at least one solution.

**Answer: E**

I and II are read from the graph of  $\sin x$  over one period. III is false:  $\sin x = 2$  has no solutions.

**Full method - how and why**

Read the claims off the graph of  $y = \sin x$  over one period, then remember the range.

- 1 I: for  $-1 < k < 1$  the horizontal line  $y = k$  cuts one period of the sine curve exactly twice. **True.**  
why: Sine rises from 0 to 1, falls to  $-1$  and returns - a strictly-between value is hit once on the way down and once elsewhere in  $[0, 2\pi)$ .
- 2 II:  $k = 1$  is reached only at the single peak  $x = \frac{\pi}{2}$ . **True.**  
why: The maximum is attained once per period.
- 3 III: for  $|k| > 1$  there is no solution at all -  $\sin x = 2$  is impossible. **False.**  
why: The range of sine is  $[-1, 1]$ .
- 4 So I and II.  
why: Match to the list.

**Where it goes wrong**

- Accepting III by forgetting the range restriction.
- Rejecting II by counting  $x = \frac{\pi}{2}$  and  $x = \frac{5\pi}{2}$  - the second is outside  $0 \leq x < 2\pi$ .
- Rejecting I because  $k = 0$  feels special:  $\sin x = 0$  gives  $x = 0$  and  $x = \pi$  in the interval - still exactly two.

**Takeaway** Per period,  $\sin x = k$  has two solutions for  $|k| < 1$ , one for  $|k| = 1$  and none for  $|k| > 1$ . Sketch the line across the curve.

### Question 11 X2-06 · Exponentials and logarithms · spec MM5.2

Which of the following is a counterexample to the claim 'for all positive  $x$  and  $y$ ,  $\log_{10}(x + y) \leq \log_{10} x + \log_{10} y$ '?

**Answer: E**

$x = y = 1$ :  $\log 2 > 0 = \log 1 + \log 1$ , so the inequality fails. For the others  $x + y \leq xy$  and the claim holds (log is increasing).

#### Full method - how and why

To break an inequality claim, look for the case where the two sides are closest - here small values, where logs go to zero or negative.

1 Test  $x = y = 1$ : left side  $\log_{10} 2 \approx 0.30$ ; right side  $\log_{10} 1 + \log_{10} 1 = 0$ .

why:  $\log 1 = 0$  for any base, so the right side collapses.

2  $0.30 > 0$ , so the claimed inequality  $\leq$  fails. **Counterexample.**

why: Hypothesis satisfied ( $x, y$  positive), conclusion false.

3 Why the others work:  $\log x + \log y = \log(xy)$ , and log is increasing, so the claim is just  $x + y \leq xy$ . For  $x = y = 2$ :  $4 \leq 4$  ✓;  $x = y = 4$ :  $8 \leq 16$  ✓;  $x = y = 5$ :  $10 \leq 25$  ✓;  $x = 10, y = 100$ :  $110 \leq 1000$  ✓.

why: Converting the log inequality into an algebraic one makes all five cases quick to check.

#### Where it goes wrong

- Trying large values, where  $xy$  dwarfs  $x + y$  and the claim comfortably holds.
- Forgetting  $\log_{10} 1 = 0$  and miscomputing the right side.
- Thinking  $\log(x + y) = \log x + \log y$  - that is the error the whole question is built on.

**Takeaway**  $\log x + \log y = \log(xy)$ , never  $\log(x + y)$ . Test inequality claims near the boundary, where the sides are closest.

**Question 12** GRA-15 · Graphs of functions · spec MM8.6

The cubic  $y = x^3 + ax + b$  has three distinct real roots. Which of the following statements must be true?

- I  $a < 0$
- II  $b \neq 0$
- III The curve has two turning points.

**Answer: F**

Three distinct real roots require two turning points (III), which needs  $y' = 3x^2 + a$  to have two roots, so  $a < 0$  (I). II is false:  $x^3 - x$  has roots  $0, \pm 1$  and  $b = 0$ .

**Full method - how and why**

Conditions for a cubic to have three distinct roots. Two claims follow from the shape; the third needs a counterexample.

- 1 III: three distinct real roots force the curve to turn twice (up, down, up), so it must have two turning points.

**True.**

why: Between consecutive roots the curve must turn around.

- 2 I:  $y' = 3x^2 + a$  has two real roots only when  $a < 0$  (so that  $x^2 = -\frac{a}{3}$  has solutions). **True.**

why: This follows directly from III: two turning points require  $y'$  to have two roots.

- 3 II: test  $y = x^3 - x = x(x - 1)(x + 1)$ : three distinct roots  $0, \pm 1$ , and here  $b = 0$ . **False.**

why: One counterexample is enough;  $b = 0$  simply means the origin is a root.

- 4 I and III only.

why: Match to the options.

**Where it goes wrong**

- Assuming  $b \neq 0$  because 'otherwise the cubic factorises' - it does, and that is fine.
- Failing to link I to III and guessing the sign of  $a$ .

**Takeaway** Three distinct roots  $\Leftrightarrow$  two turning points with values of opposite sign; that forces  $a < 0$  but says nothing about  $b$ .

**Question 13** RSN-09 · Chains of reasoning · spec Prf5

A rectangle has integer side lengths and its area (in  $\text{cm}^2$ ) is numerically equal to its perimeter (in cm). How many such rectangles are there (counting  $a \times b$  and  $b \times a$  once)?

**Answer: E**

$$ab = 2a + 2b \Rightarrow (a - 2)(b - 2) = 4: (a - 2, b - 2) = (1, 4) \text{ or } (2, 2), \text{ giving } 3 \times 6 \text{ and } 4 \times 4.$$

**Full method - how and why**

Turn a two-variable Diophantine equation into a product by the 'add a constant and factorise' trick.

- 1 Set up:  $ab = 2a + 2b$  with  $a, b$  positive integers.

why: Area equals perimeter, numerically.

- 2 Rearrange:  $ab - 2a - 2b = 0$ , then add 4 to both sides:  $ab - 2a - 2b + 4 = 4$ .

why: The added constant is the product of the two subtracted coefficients, chosen to make the left side factorise.

- 3 Factorise:  $(a - 2)(b - 2) = 4$ .

why: Check by expanding:  $ab - 2a - 2b + 4$  ✓.

- 4 Positive factor pairs of 4 with  $a - 2 \leq b - 2$ : (1, 4) and (2, 2), giving  $3 \times 6$  and  $4 \times 4$ .

why: Negative factors would need  $a < 2$ ;  $a = 1$  gives  $b - 2 = -4$ , i.e.  $b = -2$ , not a length.

- 5 So **two** rectangles.

why: Check:  $3 \times 6$  has area 18 and perimeter 18 ✓;  $4 \times 4$  has 16 and 16 ✓.

**Where it goes wrong**

- Counting  $3 \times 6$  and  $6 \times 3$  separately - the question says to count each rectangle once.
- Forgetting to add 4 and being unable to factorise.
- Including the factor pair (4, 1) as a third solution.

**Takeaway**  $ab + pa + qb = r$  factorises after adding  $pq$ :  $(a + q)(b + p) = r + pq$ . Then list factor pairs.

**Question 14** TRG-05 · Trigonometric functions and equations · spec MM4.4

Which of the following statements about the graph of  $y = \cos x$  is/are true?

I It is symmetric about the y-axis.

II It is a translation of the graph of  $y = \sin x$  by  $\frac{\pi}{2}$  in the negative  $x$ -direction.

III It has period  $\pi$ .

**Answer: E**

$$\cos(-x) = \cos x \text{ so I is true. } \sin(x + \frac{\pi}{2}) = \cos x, \text{ so II is true. The period of } \cos \text{ is } 2\pi, \text{ so III is false.}$$

**Full method - how and why**

Three properties of the cosine graph. Each is a standard fact; the false one is about the period.

- 1 I:  $\cos(-x) = \cos x$ , so the graph is unchanged by reflection in the y-axis. **True.**

why: Cosine is an even function - its graph is symmetric about the y-axis, unlike sine.

- 2 II:  $\sin(x + \frac{\pi}{2}) = \cos x$ , and replacing  $x$  by  $x + \frac{\pi}{2}$  shifts the graph  $\frac{\pi}{2}$  to the left. **True.**

why: Check a point:  $\sin$  at  $x = \frac{\pi}{2}$  equals 1, and  $\cos$  reaches 1 at  $x = 0$  - shifted left by  $\frac{\pi}{2}$  ✓.

- 3 III: the period of  $\cos$  is  $2\pi$ , not  $\pi$ . **False.**

why:  $\cos(x + \pi) = -\cos x$ , so a shift of  $\pi$  flips the graph rather than repeating it.

- 4 I and II only.

why: Match to the option list.

**Where it goes wrong**

- Thinking the period is  $\pi$  because  $\cos$  returns to zero every  $\pi$  - the **whole** shape must repeat.
- Getting the direction of the shift in II backwards.

**Takeaway**  $\cos$  is even with period  $2\pi$ ;  $\cos x = \sin(x + \frac{\pi}{2})$  is a shift to the left.

### Question 15 RSN-15 · Chains of reasoning · spec Prf5

A sequence of 10 integers is such that the sum of any three consecutive terms is positive and the sum of any five consecutive terms is negative. Which of the following statements must be true?

- I The sum of all 10 terms is negative.
- II The sequence contains at least one negative term.
- III The sequence contains at least one positive term.

**Answer: H**

Terms 1-5 and 6-10 are two blocks of five, each with negative sum (I). A five-block has negative sum, so some term is negative (II). A three-block has positive sum, so some term is positive (III).

#### Full method - how and why

Group the terms into blocks that the given conditions control, then read off each claim.

- 1 I: split the ten terms into terms 1-5 and terms 6-10 - two blocks of five consecutive terms.  
*why:* Each block has negative sum by hypothesis, and the two blocks together are the whole sequence.
- 2 Negative + negative = negative, so the total is negative. **True.**  
*why:* The grouping is exact - no term is used twice or left out.
- 3 II: terms 1-5 sum to a negative number, so at least one of them is negative (five non-negative terms could not). **True.**  
*why:* A sum is negative only if some summand is.
- 4 III: terms 1-3 sum to a positive number, so at least one is positive. **True.**  
*why:* Same argument the other way round.
- 5 So all three.  
*why:* Match to the list. (Such sequences do exist for 10 terms - the conditions become contradictory only for longer sequences.)

#### Where it goes wrong

- Trying to construct an explicit sequence; the claims follow from the block structure alone.
- Thinking the sum of all ten is ambiguous because 10 is not a multiple of 3 - use the blocks of five, which do fit.

**Takeaway** Partition the sequence into blocks the hypotheses control; a negative sum forces a negative term, a positive sum a positive one.

### Question 16 LGC-16 · The logic of arguments · spec Arg3

Which of the following statements is/are true about real numbers?

- I For all  $x$  and all  $y$ , if  $x < y$  then  $x^2 < y^2$ .
- II For all  $x$  there exists  $y$  such that  $x < y$  and  $x^2 < y^2$ .
- III There exist  $x$  and  $y$  with  $x < y$  and  $x^2 > y^2$ .

**Answer: G**

I fails for  $x = -2$ ,  $y = 1$ . II: take  $y = |x| + 1$ . III:  $x = -2$ ,  $y = 1$  works.

#### Full method - how and why

Three quantified statements about squaring and ordering. Negative numbers decide all three.

- 1 I:  $x < y \Rightarrow x^2 < y^2$  for all reals? Take  $x = -2$ ,  $y = 1$ :  $-2 < 1$  but  $4 > 1$ . **False.**  
why: Squaring is only order-preserving for non-negative numbers.
- 2 II: for each  $x$ , is there  $y$  with  $x < y$  and  $x^2 < y^2$ ? Take  $y = |x| + 1$ : then  $y > x$  and  $y > |x|$  so  $y^2 > x^2$ . **True.**  
why: Choosing  $y$  larger than  $|x|$  guarantees both conditions at once - and  $y$  may depend on  $x$ .
- 3 III: do some  $x < y$  have  $x^2 > y^2$ ? The counterexample from I works:  $x = -2$ ,  $y = 1$ . **True.**  
why: Reuse the work - the same pair settles both.
- 4 So II and III.  
why: Match to the list.

#### Where it goes wrong

- Accepting I by testing only positive numbers.
- Rejecting II by picking a bad  $y$ ; you get to choose  $y$ , so choose a helpful one.

**Takeaway**  $x < y \Rightarrow x^2 < y^2$  needs  $0 \leq x$ ; and to prove  $\forall x \exists y$ , construct  $y$  as a formula in  $x$ .

**Question 17** QUA-17 · Quadratic functions · spec MM1.3

The quadratic equation  $ax^2 + bx + c = 0$  (with  $a \neq 0$ ) has real coefficients. Which of the following statements is/are always true?

- I If  $ac < 0$ , the equation has two distinct real roots.
- II If  $b = 0$ , the equation has no real roots.
- III If the equation has a repeated root, then  $b^2 = 4ac$ .

**Answer: F**

I:  $b^2 - 4ac > 0$  whenever  $ac < 0$ , so true. II:  $x^2 - 4 = 0$  has  $b = 0$  and real roots, so false. III: a repeated root means the discriminant is zero, so true.

**Full method - how and why**

Three general claims about  $ax^2 + bx + c = 0$ . Each is a discriminant statement in disguise; the false one needs a counterexample.

- 1 I: if  $ac < 0$  then  $-4ac > 0$ , so  $b^2 - 4ac \geq -4ac > 0$ : two distinct real roots. **True.**  
*why:*  $b^2 \geq 0$  always, so a positive  $-4ac$  forces the whole discriminant positive. Geometrically,  $a$  and  $c$  of opposite signs means the curve crosses the axis.
- 2 II: test  $b = 0$  with  $x^2 - 4 = 0$ , which has roots  $\pm 2$ . **False.**  
*why:* One counterexample settles a 'for all' claim. With  $b = 0$  the discriminant is  $-4ac$ , whose sign depends on  $a$  and  $c$ .
- 3 III: a repeated root means the discriminant is zero, i.e.  $b^2 - 4ac = 0$ , i.e.  $b^2 = 4ac$ . **True.**  
*why:* This is the definition of the repeated-root case, just rearranged.
- 4 I and III only.  
*why:* Match the verdicts to the options.

**Where it goes wrong**

- Believing II because  $b = 0$  'removes' the linear term and the graph 'sits on the axis' - it does not;  $x^2 - 4$  is a counterexample.
- Mis-stating III as  $b^2 = -4ac$ .

**Takeaway** Statements about roots translate into statements about  $b^2 - 4ac$ ; to refute one, look for the simplest numbers that fit the hypothesis.

### Question 18 PRF-10 · Mathematical proof · spec Prf1

Consider the claim: **if  $x$  and  $y$  are real numbers with  $x + y > 10$ , then  $x > 5$  or  $y > 5$** . Which of the following proves it?

**Answer: A**

Supposing  $x \leq 5$  and  $y \leq 5$  and deriving  $x + y \leq 10$  is a valid proof by contradiction. 'If  $x > 5$  then the conclusion holds' assumes the conclusion;  $x = y = 6$  is a single example; 'suppose  $x > 5$ , then  $y > 5$ ' proves something false; and the averaging argument ends with 'both exceed 5', which is wrong.

#### Full method - how and why

Proving 'if  $P$  then  $(A \text{ or } B)$ '. Assuming both  $A$  and  $B$  fail and deriving a contradiction is the natural route.

- 1 Negate the conclusion: 'not  $(x > 5 \text{ or } y > 5)$ ' means  $x \leq 5$  **and**  $y \leq 5$  (De Morgan).  
why: Negating an 'or' gives an 'and' - and it gives you two usable facts instead of a disjunction.
- 2 Add the inequalities:  $x + y \leq 10$ .  
why: Inequalities of the same direction may be added.
- 3 That contradicts the hypothesis  $x + y > 10$ , so the conclusion must hold.  
why: Hypothesis true and conclusion false is impossible, which is exactly what proves the implication.

#### Where it goes wrong

- 'If  $x > 5$  then the conclusion holds' - this assumes part of what is to be proved.
- Testing  $x = y = 6$  - a single example proves nothing about all  $x, y$ .
- 'Suppose  $x > 5$ , then  $y > 10 - x$ , so  $y > 5$ ' - the algebra is wrong and it proves a false statement (both exceeding 5).
- 'The average exceeds 5 so both exceed 5' - the average exceeding 5 only forces **at least one** to.

**Takeaway** To prove an 'or', negate it into an 'and' and look for a contradiction - De Morgan turns a weak conclusion into strong assumptions.

### Question 19 LGC-20 · The logic of arguments · spec Arg1

Which of the following statements is/are logically equivalent to '**if it is raining, then the ground is wet**'?

- I It is not raining, or the ground is wet.
- II If the ground is not wet, then it is not raining.
- III If the ground is wet, then it is raining.

**Answer: E**

' $A \Rightarrow B$ ' is equivalent to 'not  $A$ , or  $B$ ' (I) and to its contrapositive (II). III is the converse, which is not equivalent.

#### Full method - how and why

Two standard equivalents of  $A \Rightarrow B$ : the contrapositive, and the disjunctive form 'not  $A$ , or  $B$ '.

- 1 I: 'not  $A$  or  $B$ '. An implication only fails when  $A$  is true and  $B$  false - exactly when ' $\neg A$  or  $B$ ' fails. **Equivalent**.  
why: Check the one bad case: raining and ground dry makes both statements false; every other case makes both true.
- 2 II: 'if the ground is not wet, it is not raining' - the contrapositive. **Equivalent**.  
why: Swap and negate.
- 3 III: 'if the ground is wet, it is raining' - the converse. **Not equivalent**; the ground could be wet from a burst pipe.  
why: A concrete alternative cause is the cleanest refutation.
- 4 So I and II.  
why: Match to the list.

#### Where it goes wrong

- Rejecting I because it looks nothing like an implication - test it against the single case that breaks an implication.
- Accepting III because wet ground is evidence of rain; evidence is not implication.

**Takeaway**  $A \Rightarrow B$  is equivalent both to 'not  $A$ , or  $B$ ' and to  $\neg B \Rightarrow \neg A$ . The converse is never among the equivalents.

## Question 20 X2-02 · Differentiation · spec MM6.3

Let  $f$  be a polynomial.  $P$ : ' $f'(x) > 0$  for all real  $x$ '.  $Q$ : ' $f(x) = 0$  has exactly one real root'. Which one of the following is true?

**Answer: C**

If  $f' > 0$  everywhere,  $f$  is strictly increasing and (being an odd-degree polynomial) crosses the axis exactly once:  $P$  is sufficient. But  $f(x) = x^3$  has exactly one root and  $f'(0) = 0$ , so  $P$  is not necessary.

### Full method - how and why

A necessary/sufficient question about polynomials. Test each direction, and look for a boundary case to break the converse.

- 1 Sufficiency: if  $f'(x) > 0$  for all  $x$ , then  $f$  is strictly increasing, so it takes each value at most once - at most one root.  
why: A strictly increasing function cannot return to a value it has already passed.
- 2 Moreover such an  $f$  must have odd degree (an even-degree polynomial turns), so it goes from  $-\infty$  to  $+\infty$  and crosses the axis. Exactly one root:  $P \Rightarrow Q$ . **Sufficient.**  
why: Both halves are needed: 'at most one' plus 'at least one'.
- 3 Necessity: find an  $f$  with exactly one root but  $f'$  not always positive. Take  $f(x) = x^3$ : one root at 0, but  $f'(0) = 0$ .  
why: A stationary point of inflexion satisfies  $Q$  without satisfying  $P$ .
- 4 So  $P$  is sufficient but not necessary.  
why: Match to the list.

### Where it goes wrong

- Claiming  $P$  is necessary because 'one root suggests increasing' -  $x^3$  and also  $x^2$ -type shifts break that.
- Forgetting that  $f' > 0$  forces odd degree, so 'at least one root' still needs justifying.

**Takeaway** Strictly increasing  $\Rightarrow$  at most one root; but a single root does not force a strictly positive derivative ( $x^3$ ).